**Theorem III.** *The forms taken over from the binary domain, that is, those forms which arise from the system forms of the corresponding binary form by bordering according to Clebsch's transfer principle, form a relatively complete system modulo* $(abc)$.

**Proof.** To a binary relation asserting that the forms $Q'_1, \dots, Q'_n$ form a complete system of a binary ground form,

$$Q' - F(Q') = 0 = \sum_{\lambda} \bigl\{(ab)c_x + (bc)a_x + (ca)b_x\bigr\}\varphi^*_\lambda,$$

there corresponds, by bordering, the ternary relation

$$Q - F(Q_i) = \sum_{\lambda} u_x \cdot (abc)\varphi_\lambda.$$

This is precisely the defining equation for a relatively complete system modulo $(abc)$. For the biquadratic form, the system of the forms taken over consists of the following forms:

$$f = a_x^4, \qquad \theta = \theta_x^4 u_\vartheta^2 = (abu)^2 a_x^2 b_x^2, \qquad K = K_x^6 u_k^3 = (a\theta u) u_\vartheta^2 a_x^3 \theta_x^3,$$
$$j = u_j^6 = (a\theta u)^4 u_\vartheta^2, \qquad \nu = u_\nu^4 = (abu)^4,$$

among which the first four form a relatively complete system modulo $((abc), \nu)$.

## § 5. Reduction of the modulus $(abc)$ to the moduli $\mathcal{A}$ and $\nu$.

The modulus $(abc)$, given only by a bracket factor, is to be reduced, in agreement with the definition (§ 3, a), to forms of the system. In other words, the form sequence $(abc)$ is to be normalized uniquely (cf. § 1).

$$a_s^2 a_x u_x = \frac{1}{3} a_s^3 u_x = \frac{1}{3} S \cdot u_x. \tag{2}$$

$$\left.\begin{aligned} (a\mathcal{A}u)^2 &= a_s - \frac{S}{6} u_x^2, \\ (a\mathcal{A}u)^3 &= 0. \end{aligned}\right\} \tag{3}$$

$$\left.\begin{aligned} \theta(s) &= (ss,x)^2 = 2a_t + \frac{1}{3} S \cdot \theta - \frac{1}{3} T u_x^2, \\ \theta(t) &= (tt,x)^2 = -\frac{1}{3} S \cdot a_t + \frac{2}{3} T \cdot a_s + \frac{1}{18} S^2 \cdot \theta - \frac{1}{18} u_x^2 \cdot S \cdot T. \end{aligned}\right\} \tag{4}$$

Sequence of moduli: $(abc)$; $\mathcal{A}$; $s$; $t$ (the system of the modulus $t$ consists of the single form $t$).[1]

It will be shown that the forms

$$\mathcal{A} = (abc)^2 a_x^2 b_x^2 c_x^2, \qquad \nu = (abu)^4$$

suffice as moduli, that is, that the forms

$$\mathcal{A} = (abc)^2 a_x^2 b_x^2 c_x^2, \qquad a_\nu = (abc)(bcu)^3 a_x^3,$$
$$a_\nu^2 = (abc)^2 (bcu)^2 u_x^2, \qquad i = a_\nu^4 = (abc)^4$$

define the form sequence $(abc)$. In the form sequence $a_\nu$, the form $a_\nu^3$ is missing according to the identity

$$a_\nu^3 = (abc)^3 a_x (bcu) = \frac{1}{3} u_x \cdot (abc)^4 = \frac{1}{3} i \cdot u_x.$$

[1] Gordan–Kerschensteiner, loc. cit., p. 134.

For reducing a modulus to a higher one, according to the definition, we have to reduce the most general foldings, or all special foldings coming into consideration, with the modulus to foldings with the higher modulus.

In the present case the most general foldings arise from the expressions

$$(abc)a_x^3b_y^3c_z^3, \qquad (abc)a_x^2b_y^2c_z^2a_zb_xc_x$$

(taken over from the cubic forms),

$$(abc)a_xb_yc_za_x^2b_x^2c_x^2$$

(taken over from the quadratic forms), and from the polarisation of these expressions.

For the calculation of these expressions by the polars of $\mathcal{A}$ and of the form sequence $a_\nu$, respectively of their initial terms, we take the indirect route, as in § 2. We expand the polar terms

$$(abc)^2a_x^2b_y^2c_z^2, \qquad a_\nu(\nu xy)(\nu yz)(\nu zx)a_xa_ya_z = (abc)(bc\widehat{x}u)(bc\widehat{y}z)(bc\widehat{z}x)a_xa_ya_z \quad \text{etc.}$$

as linear aggregates of expressions with the symbolic factor $(abc)$ and obtain, by inverting the system of equations, the desired relations, as well as a relation between the polars taken according to different combinations of the variables. The identity theorem and the product theorem for determinants are used in the evaluation.

The system of equations is, for $(abc)a_x^3b_y^3c_z^3$:

$$1)\quad \sum_3 a_\nu(\nu yz)^3a_x^3 = 6(abc)a_x^3b_y^3c_z^3 - 6\sum_3(abc)a_x^3b_y^2b_zc_z^2c_y^*,$$

$$\begin{aligned}2)\quad 3(abc)^2a_x^2b_y^2c_z^2\cdot(xyz) - \frac{1}{2}\sum_3 a_\nu^2(\nu yz)^2\nu_x^2(xyz) &= 2\sum_3(abc)a_x^3b_y^2b_zc_z^2c_y\\ &\quad - 4\sum_3(abc)a_xa_ya_zb_y^2b_zc_z^2c_x,\end{aligned}$$

$$3)\quad a_\nu(\nu xy)(\nu yz)(\nu zx)a_xa_ya_z = 2\sum_3(abc)a_xa_ya_zb_y^2b_zc_z^2c_x,$$

$$4)\quad \sum_3 a_\nu(\nu yz)^3u_x^3 - \frac{3}{2}\sum_3 a_\nu^2(\nu yz)^2\nu_x^2(xyz) + \frac{1}{6}i\cdot(xyz)^3 = 6\sum_3(abc)a_xa_ya_zb_y^2b_zc_z^2c_x.$$

It follows for

$$(abc)a_x^3b_y^3c_z^3,$$

and by analogous calculation for

$$(abc)a_x^2b_y^2c_z^2a_zb_xc_x, \qquad (abc)a_xb_yc_za_z^2b_x^2c_x^2:$$

$$\begin{aligned}(abc)a_x^3b_y^3c_z^3 &= \frac{3}{2}(abc)^2a_x^2b_y^2c_z^2(xyz) + \frac{3}{2}a_\nu(\nu xy)(\nu yz)(\nu zx)a_xa_ya_z\\ &\quad - \frac{1}{12}i\cdot(xyz)^3,\end{aligned}$$

$$\text{(IV.)}\quad \begin{aligned}(abc)a_x^2b_y^2c_z^2a_zb_xc_x &= \frac{2}{3}(abc)^2a_x^2b_y^2c_z^2c_x\cdot(xyz)\\ &\quad + \frac{1}{3}a_\nu(\nu xy)(\nu yz)(\nu zx)a_x^3 - \frac{1}{6}a_\nu^2(\nu xy)(\nu zx)a_x^2\cdot(xyz),\end{aligned}$$

$$(abc)a_xb_yc_za_z^2b_x^2c_x^2 = \frac{1}{6}(abc)^2a_x^2b_z^2c_z^2\cdot(xyz).$$

We have therefore obtained a relatively complete system modulo $(\mathcal{A},\nu)$, consisting of the four forms $f,\theta,K,j$. It follows from this that all transvections of $f$ over $\theta$ not taken over from the binary domain, as well as the transvection $(a\theta u)^2$ reducible in the binary case to $(ab)^4$, can be expressed by the symbols $\mathcal{A}$ and $\nu$.[2]

We obtain the reduction formulas fundamental for what follows by specializing expansion IV, or more shortly by direct calculation (interchanging the symbols), for $a_\vartheta(a\theta u)^2$, $a_\vartheta^2((a\theta u)^2$, $(a\theta u)^2$, according to the following ansatz:

$$\begin{aligned} a_\vartheta(a\theta u)^2 &= (abc)(bcu)(abu)^2 - \frac{1}{3}(abc)(bcu)\{(bcu)a_x - u_x(abc)\}^2, \\ a_\vartheta^2(a\theta u)^2 &= (abc)^2(abu)^2 - \frac{1}{3}(abc)^2\{(bcu)a_x - u_x(abc)\}^2, \end{aligned}$$

for $((a\theta u)^2)^*$:

$$\begin{aligned} (abu)a_y^3b_x^3 &= \frac{3}{2}u_\vartheta(\vartheta yx)\theta_y^2 + \frac{1}{4}(\nu yx)^3, \\ ((abu)(acu))b_x^3 &= \frac{3}{2}u_\vartheta(\vartheta\widehat{a}ux)(\theta au)^2 + \frac{1}{4}(\nu\widehat{a}ux)^3. \end{aligned}$$

$$\left.\begin{aligned} (a\theta u)^2 &= \frac{1}{6}\nu\cdot f - \frac{2}{3}u_x\cdot a_\nu + \frac{2}{3}u_x^2\cdot a_\nu^2 - \frac{i}{18}u_x^4, \\ a_\vartheta &= \frac{1}{3}u_x\cdot\mathcal{A}, \\ a_\vartheta(a\theta u) &= 0, \\ a_\vartheta^2 &= \mathcal{A}, \\ (a\theta u)^3 &= 0, \\ a_\vartheta(a\theta u)^2 &= -\frac{5}{6}a_\nu + \frac{7}{6}u_x\cdot a_\nu^2 - \frac{i}{9}u_x^3, \\ a_\vartheta^2(a\theta u) &= 0, \\ (a\theta u)^4 &= j, \\ a_\vartheta(a\theta u)^3 &= 0, \\ a_\vartheta^2(a\theta u)^2 &= \frac{2}{3}a_\nu^2 - \frac{i}{9}u_x^2. \end{aligned}\right\} \tag{1}$$

We add the formula, likewise obtained from the reduction of the modulus $(abc)$:

$$a_\nu^3a_xu_\nu = \frac{1}{3}i\cdot u_x. \tag{2}$$

*From formulas (1.) and (2.) and from the formulas formed analogously for the ground form $\nu$, all later reduction formulas are derived by polarisation*, except for the relations for the decomposed forms. From formulas (1.) we see:

The form sequence leads:

$$\begin{array}{rll} a_\vartheta(a\theta u)^2 & \text{to} & \text{symbols } \nu \text{ alone}, \\ a_\vartheta & \text{to} & \text{symbols } \mathcal{A} \text{ and } \nu, \\ (a\theta u)^2 & \text{to} & \text{symbols } j,\mathcal{A} \text{ and } \nu, \\ (a\theta u)^2 \text{ under folding I} & \text{to} & \text{symbols } j \text{ and } \nu, \\ (a\theta u)^2 \text{ under folding II} & \text{to} & \text{symbols } \mathcal{A} \text{ and } \nu. \end{array}$$

We can therefore replace:

[2] $\sum_3$ refers to the sum of three terms obtained from the initial term by cyclic interchange of the variables. The equations also hold individually for each term of the sum.

1. modulo $(\mathcal{A}, \nu)$: foldings with $j$ are replaced by corresponding foldings with $(a\theta u)^2$ or $(a\theta u)^3$;

2. modulo $(\nu)$: foldings with $\mathcal{A}$ are replaced by corresponding foldings with $a_\vartheta$ or $a_\vartheta(a\theta u)$, or also by special foldings with $(a\theta u)^2$ (which lead only to a single folding I).

In particular:

$$(a\theta u)^2\theta_y^2\theta_x^2 = \frac{1}{3}(jyx)^2 \pmod{\nu}, \qquad (a\theta u)^2 u_y\theta_y = -\frac{1}{6}(jyx)^2 \pmod{\nu},$$
$$(a\theta u)^2\nu_y^2 = \frac{1}{3}(\mathcal{A}u\nu)^2 \pmod{\nu}, \qquad (a\theta u)(a\theta\nu)u_\vartheta\nu_\vartheta = -\frac{1}{6}(\mathcal{A}u\nu)^2 \pmod{\nu},$$
$$(a\theta u)^2u_\nu^2\vartheta_\nu^2 = \frac{1}{3}(\mathcal{A}u\nu)^2\mathcal{A}_\nu \pmod{\nu}.$$

## § 6. Reduction of the modulus $(\mathcal{A}, \nu)$ to the modulus $(\nu)$.

The reduction formulas of the preceding paragraph give us the means to set up the relatively complete system modulo $\nu$ directly; we shall see that to the system of the transferred forms one has only to add the forms $\mathcal{A}$ and $(a\mathcal{A}u) = N$ (analogously as for the cubic form, and indeed valid in general).

For the reduction of the modulus $\mathcal{A}$ we consider all special foldings coming into consideration, that is, the foldings of the relatively complete system modulo $(\mathcal{A}, \nu)$ with the system of $\mathcal{A}$, and begin with the forms lowest in the coefficients, the foldings of $f$ with $\mathcal{A}$.

For the reduction of the forms $(a\mathcal{A}u)^2$, $(a\mathcal{A}u)^3$, $(a\mathcal{A}u)^4$, we replace the symbols $\mathcal{A}$ by lower forms of the form sequence according to § 5; that is, we develop the expressions

$$a_\vartheta b_\vartheta(b\theta u)^2, \qquad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2, \qquad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3$$

1) by polars of the form sequence $a_\vartheta$ (symbols $\mathcal{A}$ and $\nu$),

2) by polars of the form sequence $b_\vartheta(b\theta u)^2$ (symbols $\nu$),

and obtain the reduction formulas (calculation below):

$$\begin{aligned}
&\text{a)} \quad (a\mathcal{A}u)^2 = \frac{1}{2}(\vartheta\nu x)^2 - \frac{2}{5}f\cdot a_\nu^2 - \frac{4}{5}u_x\cdot\theta_\nu(\vartheta\nu x)^2 \\
&\qquad\qquad + u_x^2\left\{\frac{1}{6}i\cdot f - \frac{1}{40}S\right\}, \\
&\text{b)} \quad (a\mathcal{A}u)^3 = -\frac{3}{10}\theta_\nu(\vartheta\nu x), \\
&\text{c)} \quad (a\mathcal{A}u)^4 = \frac{21}{10}\theta_\nu^2 - \frac{3}{10}\Pi - \frac{6}{5}u_x\cdot\theta_\nu^3 + \frac{1}{10}u_x^2\cdot\theta.
\end{aligned} \tag{3}$$

(The same results are also reached by the double reduction of the expressions $a_\vartheta^2(b\theta u)^2$, $a_\vartheta^2(b\theta u)^3$, $a_\vartheta^2((a\theta u)^2(b\theta u)^2$ and $a_\vartheta^2((a\theta u)(b\theta u))^3$ after elimination of $a_\vartheta^2$.)

From formulas (3.) it follows that $(a\mathcal{A}u)^2$ is a reducer with respect to the modulus $\nu$. Hence:

1) The reduction of the system of $\mathcal{A}$: the form sequence $(\mathcal{A}\mathcal{A}'u)^2 = a_\vartheta^2((a\mathcal{A}u)^2$ has the reducer $(a\mathcal{A}u)^2$ as a factor.

2) The reduction of the system arising by folding with the modulus $\mathcal{A}$:

The form sequence $(\theta\mathcal{A}u)^2$ has the reducer $(a\mathcal{A}u)^2$ as a factor.

For the forms $(\theta\mathcal{A}u)$ and $\mathcal{A}_\vartheta$ one obtains:

$$(a\mathcal{A}u)^2(abu) = (\theta\mathcal{A}u) - u_x\cdot\mathcal{A}_\vartheta(\theta\mathcal{A}u),$$

$$(a\mathcal{A}u)(a\mathcal{A}b) = -\frac{1}{2}\mathcal{A}_\vartheta + \frac{1}{2}u_x \cdot \mathcal{A}_\vartheta^2.$$

From the reducer $(\theta\mathcal{A}u)$, however, follows the reduction of the system arising by folding with the modulus $\mathcal{A}$.